S4 Supplementary Clinical Summary

A cautious clinician-facing interpretation of the occupancy result

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Short title: S4 Clinical - Apollo propofol occupancy

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Scope and intent

> **Important: healthy-volunteer sedation - not a monitoring study**
>
> This is a secondary analysis of published EEG from 20 healthy adults sedated with propofol in a laboratory protocol. No clinical intervention was tested. No individual-level prediction is supported. Occupancy is not a depth-of-anaesthesia index, not a consciousness biomarker, and not a tool for titration.

This supplement is for clinically oriented readers. The goal is translation without amplification. The sample is healthy volunteers, the design is within-subject observational across protocol phases, and the main metric is a research occupancy scalar in a baseline-fixed spectral chart. It was selected post hoc, so the reported contrasts are exploratory rather than confirmatory.

What is the framework, and why does it matter here?

Anaesthesia research often asks whether the brain has “less capacity” or is “less accessible” under propofol. Those phrases mix three different facts: the measured state has moved, behaviour has changed, and the local dynamical law may or may not have changed. This article keeps those apart. Occupancy asks only whether the measured spectral pattern has left its awake neighbourhood.

What was studied

Twenty adults, four EEG sessions each: before drug, first protocol sedation, later protocol sedation, and recovery, with blood propofol levels and a 40-trial auditory discrimination task (Chennu et al., 2016). We did not collect new recordings.

What occupancy means in plain language

Occupancy is how far a participant’s current EEG spectral pattern sits from that same person’s pre-drug pattern, after a robust, within-person scaling. Larger occupancy means a larger displacement in that chart, not “more unconscious.”

The main findings, in order of clinical salience

1. Propofol moves the spectral pattern in an ordered way

Displacement increases from baseline to first sedation to later sedation, then only partly returns during recovery. This is a protocol-phase trajectory, not a clinical depth scale.

2. Exploratory responsiveness analysis

The original 7 versus 13 drowsy/responsive split is recoverable from the task scores. The exploratory data provide no evidence of a monotonic occupancy association with that split, but the impaired arm is small and this does not establish that an association is absent. Some people remain accurate despite very large displacement; some become impaired with similar or smaller displacement.

3. We cannot yet say whether local dynamics change at a comparable state

Most participants have too little overlap between awake and sedated spectral patterns for a fair “same state, different future?” test. That is a measurement boundary, not evidence that dynamics are unchanged.

What was not established

No monotonic occupancy association with behavioural responsiveness was detected in this exploratory sample, but absence of association was not established. Protocol phase 3 is not unresponsiveness. Local-chart reductions in drift/diffusion did not survive FDR and did not survive a baseline-fixed reference. No consciousness claim

is made. Feature-level contributions and three bounded scaling/feature diagnostics were assessed in a post-hoc computational branch; they supported stability of the reported scalar for those choices but are not clinical validation or evidence that other feature sets, preprocessing, or cohorts would agree.

What clinicians should not conclude

Do not use occupancy for titration, awareness monitoring, or prognosis. Do not treat a large occupancy value as proof of unresponsiveness. Do not treat incomplete occupancy return at recovery as residual anaesthesia without a behavioural or pharmacokinetic reading.

What would increase clinical confidence

Prospective lock of the occupancy scalar, larger samples with a balanced impaired arm, independent behavioural endpoints, replication of the sensitivity branch in an independent cohort, and a chart in which matched-state dynamics can actually be tested.

Summary for clinical readers

In this laboratory propofol dataset, the EEG spectral pattern reliably leaves the awake neighbourhood and only partly returns. The exploratory data provide no evidence that this displacement is a monotonic marker of behavioural responsiveness, but the analysis cannot rule out an association. Occupancy is a research description of state relocation, not a clinical marker.

Bibliography

Chennu, S., O'Connor, S., Adapa, R., Menon, D. K., & Bekinschtein, T. A. (2016). Brain Connectivity Dissociates Responsiveness from Drug Exposure during Propofol-Induced Transitions of Consciousness. *PLOS Computational Biology*, *12*(1), e1004669. https://doi.org/10.1371/journal.pcbi.1004669